



Subashini M 251801338-2025-AIDS ▾

S2

Started on	Monday, 24 August 2026, 5:19 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:30 PM
Time taken	37 days 6 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int count = 0;
8
9      int i = 1; count++;
10     int s = 1; count++;
11
12     while (s <= n) {
13         count++; // condition check s <= n (true)
14         i++; count++;
15         s += i; count++;
16     }
17     count++; // condition check s <= n (false)
18
19     printf("%d\n", count);
20
21     return 0;
22 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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